FORMAL SPECIFICATION

ERES Cryptographic Stack

GCF–EDF–USC Integration with CBGMODD×FAVORS+BERA=RATE Interlock

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Phase	Consolidation Phase → upstream of VERTECA
Governs	SECUIR cryptographic output layer
Status	**Locked and tested. Ten.**
License	CCAL v2.1 (CARE Commons Attribution License)
Validation	MIEVM (Claude, Grok, DeepSeek, ChatGPT)

1. Scope and Purpose

This specification defines the ERES Cryptographic Stack — a novel cryptographic primitive class in which authentication and authorization are achieved not through computational hardness but through resonance-keyed semantic superposition. Three polysemantic primitives (GCF, EDF, USC) each carry multiple simultaneous readings; a correct resonance signature collapses the superposition to the intended reading. Without the correct signature, the readings remain superposed and the primitive yields no operationally resolved meaning.

The stack is bound together by a single canonical interlock:

$$\text{(CBGMODD} \times \text{FAVORS)} + \text{BERA} = \text{RATE}$$

This form is structurally isomorphic to ERES Triune Math #2 (M×E+C=R). The cryptographic layer is therefore not a parallel construction but an instance of the Triune Math operating in the authentication domain.

1.1 Classification within the ERES Decision-Stack

- Consolidation Phase deliverable (not a Research Phase preliminary).
- Upstream dependency of VERTECA (voice portal semantic interpretation layer).
- Governs the cryptographic output layer of SECUIR.
- Sits alongside ERES-BODY-SPEC-2026-001 (substrate) and ERES-BRAINS-SPEC-2026-001 (cognitive organ) in the architectural core.

1.2 What This Specification Is Not

- Not a hardness-based cryptosystem. There is no trapdoor function, no discrete-log assumption, no lattice problem. Security properties derive from semantic superposition and resonance validation, not computational intractability.
- Not a replacement for transport-layer cryptography. ERES Crypto operates at the semantic and governance layer; TLS-class primitives remain appropriate for wire protection.
- Not a scoring rubric. RATE is an uncollapsed multidimensional output; it cannot be reduced to a single scalar.

2. Normative References

The following ERES documents are normatively referenced. Compliant implementations shall resolve all references to the specified version or later.

- ERES-RECKON-WP-2026-002 — Seven Resonance Anchors; conjunctive gate.
- ERES-RECKON-MATH-WP-2026-002a — ERES Triune Math canonical forms.
- ERES-BODY-SPEC-2026-001 — GAIA BODY substrate specification.
- ERES-BRAINS-SPEC-2026-001 — BRAINS Corpus Ingestion Pipeline.
- ERES TERMS #47 (ET_AL) — Locked Terms Registry; Three Governing Principles.
- ERES-COVENANT-2026-001 — ERES Equity Covenant (1000-year governance).
- ERES Talonics Symbol Matrix (452 elements, 12 parts) — notation basis for semantic encoding.

- Talonics Proof-of-Work (RG #397) — TETRA:GAIA_Primitive; BEST/SOUND/GOOD tetrahedral independence.

3. Definitions and Locked Terms

Terms in this section are locked per ERES TERMS #47 and may not be redefined within compliant implementations. Divergence constitutes a specification violation.

Term	Locked Definition
GCF	Polysemantic primitive holding three simultaneous readings: (1) Greatest Common Factor — mathematical reduction; (2) GraceChain Formula — governance formula; (3) Good Christ Family — familial-spiritual anchor. Structures merit accrual, credit conditions, and value flow.
EDF	Polysemantic primitive holding three simultaneous readings: (1) Earned Dignity Framework — personal dignity domain (ONE-GOOD); (2) Earth Defense Force — security and protection domain (SECURITY-CLEARANCE); (3) Earth Data Framework — information integrity domain (DATA-INTEGRITY). Validates resonance across three operational domains.
USC	Utterance–Sound–Cognition. 3×3 = 9 semantic dimensions. U: Unite, Universe, Us. S: Safe, Sound, Solid. C: Constant, Care, Christ. Validates expression, resonance carrier, and cognitive attestation.
CBGMODD	Scalable Certification Architecture governance operator. Four pillars: HEALTH/SSHP, LAW/SSLA, PROTECTION/SSPS, SKILLS_TRADE/SSST. Tier 1 = SSSC (Solid-State Smart-City Citizen).
FAVORS	Six-channel biometric validation operator: Fingerprint, Aura, Voice, Odor, Retina, Signature. Aura channel coupled to BERA ARI.
BERA	BERA Resonance Index: ARI (Aura Resonance Index), ERI (Emission Resonance Index), RHC (Resonant Harmony Cycle), RCI (Resonant Continuity Index).
RATE	Uncollapsed multidimensional cryptographic output. Scored 1–10 independently across enumerated dimensions (see §7). Not reducible to a single scalar.
Resonance Signature	The context-specific pattern (derived from BERA plus Talonics symbol context) that selects which reading of each polysemantic primitive is active. Functions as the cryptographic key.
Superposition	The state of a polysemantic primitive prior to resolution by a resonance signature, in which all enumerated readings coexist without privilege.
Conjunctive Gate	Per ERES-RECKON-WP-2026-002: all required inputs must be non-null for output to exist. Prevents GDP-style compensation across dimensions.

4. Polysemantic Primitives

Polysemantic primitives are the core novelty of the ERES Cryptographic Stack. Each primitive is defined as a finite ordered set of readings. The resonance signature selects the active reading; without the signature, the primitive remains in superposition.

4.1 GCF — Three-Dimensional Cipher

Channel	Reading	Operational Role
G_1	Greatest Common Factor	Mathematical reduction operator; normalizes merit claims to lowest common terms.
G_2	GraceChain Formula	Governance formula; encodes credit conditions and value flow across GraceChain.
G_3	Good Christ Family	Familial-spiritual anchor; binds merit accrual to covenantal ground.

4.2 EDF — Three-Dimensional Cipher

Channel	Reading	Operational Domain
E_1	Earned Dignity Framework	ONE-GOOD / personal dignity (Trilogy Book 1; UBIMIA-adjacent).
E_2	Earth Defense Force	SECURITY-CLEARANCE / protection (Trilogy Book 2; IDIPITIS–NBERS–SCALULAR).
E_3	Earth Data Framework	DATA-INTEGRITY / information (Trilogy Book 3; FAVORS–CBGMODD–GAIA–SOMT).

The three EDF readings map one-to-one onto the Trilogy. EDF is therefore not merely a cipher but the cryptographic form of the entire ERES canonical arc.

4.3 USC — Nine-Dimensional Cipher

USC is the largest primitive, yielding a 3×3 semantic matrix. Each column corresponds to a pipeline stage in the VERTECA voice portal: utterance production, sound propagation, cognitive attestation.

Row	U — Utterance	S — Sound	C — Cognition
1	Unite	Safe	Constant
2	Universe	Sound	Care
3	Us	Solid	Christ

USC validates the three-stage pipeline: expression is spoken (U), carried as resonance (S), and attested by cognition (C). A complete USC resolution selects one cell per column, yielding 3^3 = 27 possible triples. The resonance signature selects exactly one.

4.4 Resonance-Keyed Semantic Superposition

Formally, a polysemantic primitive P of arity n is defined as:

$$P = \{ r_1, r_2, \ldots, r_n \} \quad \text{with } \mathrm{resolve}(P, \sigma) \rightarrow r_i$$

where σ is a resonance signature and resolve() is the selection function. Absent a valid σ, P remains in superposition and no reading is operationally active. The cryptographic property is therefore: an adversary without the correct σ cannot determine which reading the sender intended, because no single reading is privileged in P itself.

This is fundamentally distinct from probabilistic encryption. A ciphertext in standard crypto has one intended plaintext hidden by hardness; a polysemantic primitive in ERES Crypto has multiple simultaneous readings, with intent selected by resonance. The attack surface is not computational but semantic: an adversary must forge a resonance signature, which requires biometric attestation (FAVORS) plus governance substrate (CBGMODD) plus matching BERA context.

5. Cryptographic Interlock Formula

5.1 Canonical Form

$$(\text{CBGMODD} \times \text{FAVORS}) + \text{BERA} = \text{RATE}$$

Parentheses are canonical. The multiplicative grouping of CBGMODD and FAVORS is mandatory. Any implementation that computes CBGMODD × (FAVORS + BERA) is non-conformant.

5.2 Semantics of Each Operator

- **× (multiplicative conjunction):** Governance substrate and biometric activation must both be non-null for the earned component to exist. Zero in either collapses the product to zero regardless of BERA. This is the conjunctive gate operating at the crypto layer.
- **+ (semantic composition):** BERA supplies the resonance signature that keys the polysemantic primitives. The + operator is arithmetic in the Triune Math isomorphism and compositional in the cryptographic reading — these are two views of the same operation.
- **= (resolution):** Produces RATE as uncollapsed multidimensional output. The equation is not an arithmetic identity reducible to a scalar — it is a resolution relation.

5.3 Component Specifications

5.3.1 CBGMODD (Governance Substrate)

CBGMODD provides certification substrate across the four SCALULAR pillars. For the interlock, CBGMODD is represented as a four-tuple:

$$\text{CBGMODD} = (\text{SSHP}, \text{SSLA}, \text{SSPS}, \text{SSST})$$

Each pillar yields a binary attestation (certified / not certified) at the active tier. CBGMODD is non-null if and only if the user's active tier (minimum SSSC for Tier 1) is certified across all four pillars required by the transaction context. Partial certification collapses CBGMODD to zero for the interlock.

5.3.2 FAVORS (Biometric Activation)

FAVORS is a six-channel biometric operator:

$$\text{FAVORS} = (\text{F}, \text{A}, \text{V}, \text{O}, \text{R}, \text{S})$$

Each channel contributes an attestation quantum. The Aura channel (A) is coupled directly to the BERA ARI reading; this coupling is the mechanism by which biometric state and resonance state are bound. FAVORS is non-null if the transaction-specific quorum of channels attests; quorum varies by context and is governed by SCALULAR tier requirements.

5.3.3 BERA (Resonance Constant)

BERA is a four-index resonance reading:

$$\text{BERA} = (\text{ARI}, \text{ERI}, \text{RHC}, \text{RCI})$$

Each index is independent. ARI and ERI measure emission state; RHC measures cyclic harmony; RCI (co-developed with Jimmy D. Butzbach's Continuity Theory) measures continuity across the cycle. BERA serves as the resonance signature σ that resolves the polysemantic

primitives. A null BERA does not zero RATE arithmetically, but leaves the polysemantic primitives in superposition — RATE is then semantically unresolvable (see §9).

6. Isomorphism to ERES Triune Math #2

The interlock formula is isomorphic to ERES Triune Math canonical form #2:

$$M \times E + C = R$$

The mapping is:

Triune Math	Crypto Interlock	Role
M (Matter / Substrate)	CBGMODD	Governance substrate — certified structure.
E (Energy / Activation)	FAVORS	Biometric activation — six-channel energetic signature.
C (Constant)	BERA	Resonance constant — context that keys the primitives.
R (Reason)	RATE	Multidimensional reason-output, uncollapsed.

This isomorphism is not metaphorical. The crypto interlock is an instance of Triune Math #2 operating in the authentication-and-authorization domain. Any future refinement of Triune Math #2 propagates automatically into the crypto layer.

7. RATE Output Specification

RATE is multidimensional and uncollapsed. Scores are integers in [1, 10] on each dimension, assigned independently. Compliant implementations shall not collapse RATE to a scalar (mean, sum, product) for any decision-critical use; such collapse reintroduces GDP-style compensation and is explicitly prohibited by ERES-RECKON-WP-2026-002.

7.1 Enumerated RATE Dimensions (v1.0)

#	Dimension	Definition
R_1	Merit Accrual Rate	Velocity at which Meritcoin accrues to the actor under the current transaction context.
R_2	Resonance Credibility	Strength and stability of the BERA-keyed resonance signature; derived from ARI·ERI·RHC·RCI coherence.
R_3	Governance Certification Speed	Speed at which CBGMODD attestations are produced and refreshed at the active tier.
R_4	Biometric Attestation Strength	Quorum margin of FAVORS channels above the transaction minimum.

#	Dimension	Definition
R_5	Semantic Clarity	Unambiguity of the resolved polysemantic reading; penalized when multiple readings remain partially active.
R_6	Temporal Persistence	Stability of the resolution over the RCI window; short-window flicker lowers this.
R_7	Three-Principle Alignment	Alignment with ERES TERMS #47: don't hurt self, don't hurt others, build for generations to come.

Version 1.0 enumerates seven dimensions. Future versions may extend this set; they shall not remove existing dimensions without a major version increment.

7.2 Reading a RATE Vector

A RATE vector is written as $\langle R_1, R_2, \ldots, R_7 \rangle$ with each $R_i \in [1, 10] \cup \{\perp\}$. The symbol $\perp$ denotes undefined and is distinct from the score 1. An undefined dimension propagates: any decision logic that encounters $\perp$ must either reject the transaction or explicitly invoke a failure branch (see §9). Silently substituting a default value for $\perp$ is a specification violation.

8. Matrix Modulation

RATE dimensions are not static. They are modulated by the set of active CBGMODD activities and are resonant-context aware. The modulation matrix M applies to the base RATE vector as:

$$\texttt{RATE_active} = \mathbf{M}(\texttt{ctx}) \circ \texttt{RATE_base}$$

where ctx is the active resonance context (pillar activity, tier level, time-of-cycle in RHC, continuity band in RCI) and $\circ$ is channel-wise modulation constrained to the [1, 10] band.

Intent is cryptographically known through the resonance signature. An actor whose BERA signature does not match the claimed intent will receive a RATE vector in which the dimensions relevant to that intent are suppressed. This is the mechanism by which ERES Crypto binds intent to authentication without requiring an explicit declaration-of-intent field.

9. Failure Taxonomy

The interlock admits exactly four principal failure modes. Each has a distinct diagnostic and a distinct required response. Implementations shall not conflate them.

Code	Condition	Cause	Required Response
F_1 GOV-NULL	CBGMODD = 0	Governance substrate absent or lapsed at required tier.	RATE undefined. Refer to NPR for remediation; do not retry.
F_2 BIO-NULL	FAVORS = 0	Biometric quorum not met; no channel substitution permitted.	RATE undefined. Re-attestation required before retry.

Code	Condition	Cause	Required Response
F_3 RES-NULL	BERA = 0	No resonance signature; polysemantic primitives remain superposed.	RATE semantically unresolvable. Distinct from arithmetic zero.
F_4 SEM-COLLIDE	Signature ambiguous	Multiple resonance signatures partially match; primitives resolve to more than one reading.	RATE rejected; escalate to MIEVM adjudication.

9.1 Why F_3 Is Not Arithmetic Zero

Naively, if BERA = 0 in (CBGMODD × FAVORS) + BERA, RATE would equal the earned component. This is the arithmetic reading, and it is correct under the Triune Math isomorphism — but it is not the cryptographic reading. Without BERA, the polysemantic primitives have no key. The earned component exists as a magnitude but has no resolved meaning. RATE is therefore semantically unresolvable, not merely reduced. Implementations shall return ⊥ for all semantically-keyed dimensions when BERA = 0, not the earned magnitude.

10. Integration with the ERES Decision-Stack

The canonical flow is: Consolidation Phase → VERTECA → PlayNAC → SECUIR. The Crypto Stack integrates at two points.

10.1 Upstream (Consolidation Phase)

- BODY (ERES-BODY-SPEC-2026-001) supplies substrate signals. ODOR in “Because Odor Does Yellow” feeds the FAVORS O channel.
- BRAINS (ERES-BRAINS-SPEC-2026-001) supplies cognitive attestation into the USC–C column.
- Talonics Symbol Matrix supplies the notation in which resonance signatures are expressed at the semantic layer.

10.2 Downstream (VERTECA → SECUIR)

- VERTECA’s voice portal uses USC as its pipeline ontology: utterance → sound → cognition.
- PlayNAC (EP GERP SOMT) consumes RATE vectors as inputs to its governance ruleset without collapsing them.
- SECUIR emits final cryptographic outputs using RATE vectors as its authorization basis.

11. Validation Requirements (MIEVM)

All non-trivial modifications to this specification shall be validated under MIEVM (Multi-Instrument Ensemble Validation Method): parallel inference across Claude, Grok, DeepSeek, and ChatGPT with adversarial roles assigned per ERES practice. The short-list is fixed at four instruments for solo-researcher bandwidth, consistent with the constraint-based epistemology that produced the 72 Key Domains.

11.1 Required Validation Passes

- Isomorphism check: the interlock must verifiably reduce to Triune Math #2 under all extensions.
- Conjunctive-gate check: no implementation path may permit GDP-style compensation among RATE dimensions.
- Failure-mode check: F_1–F_4 must be independently triggerable in test harnesses.
- Polysemantic-resolution check: primitives must return ⊥ under F_3, not the earned magnitude.
- Three-Principle check: R_7 must be queryable for every RATE vector.

12. Security Considerations

Security in ERES Crypto derives from the joint difficulty of forging three independent conditions simultaneously in context: a governance substrate (CBGMODD), a biometric quorum (FAVORS), and a live resonance signature (BERA). Each is independently hard to forge and jointly harder.

12.1 Known Threat Classes

- **Biometric replay:** Mitigated by FAVORS Aura channel coupling to live BERA ARI; a replayed biometric has stale resonance.
- **Governance impersonation:** Mitigated by multi-pillar quorum within CBGMODD; single-pillar compromise does not produce attestation.
- **Semantic collision:** Tracked as F_4; escalated to MIEVM adjudication rather than silently resolved.
- **Intent laundering:** Structurally infeasible because intent is bound to resonance signature; declared-but-unresonant intent is not authenticated.

12.2 What ERES Crypto Does Not Protect

- Wire confidentiality — use TLS-class transport.
- Coercion attacks where a legitimate actor is compelled to produce signatures under duress; this is a governance concern addressed by NPR and SECUIR, not by the crypto layer.
- Long-horizon substrate compromise where CBGMODD attestations are systematically corrupted — addressed by the ERES Equity Covenant and its 1000-year governance horizon.

13. Change Log and Provenance

Version	Date	Change
1.0	2026-04-21	Initial formal specification. Interlock locked as (CBGMODD × FAVORS) + BERA = RATE. Isomorphism to Triune Math #2 established. Seven RATE dimensions enumerated. Four-mode failure taxonomy codified.

13.1 Provenance Notes

- The decision to lock parenthetical precedence as (CBGMODD × FAVORS) + BERA was made following a precedence-ambiguity flag; the chosen form is the one that maps cleanly to Triune Math #2.
- The distinction between arithmetic zero and semantic ⊥ for BERA = 0 (F_3) was introduced in this specification as a clarification not present in the original informal statement.
- The enumeration of seven RATE dimensions is declarative for v1.0; the original informal statement mentioned “merit accrual rate, resonance credibility, governance certification speed, and other dimensions,” which are preserved here as R_1–R_3 with four additional dimensions made explicit.

14. Attestation

This specification is attested as locked and tested (“Ten”) by the author, Joseph Allen Sprute (ERES Maestro, JAS), on behalf of the ERES Institute for New Age Cybernetics. It supersedes no prior specification and is upstream of VERTECA.

— End of Specification —